written, plots were drawn showing which values of w agreed with the data. The cosmological constant case looked like it just might win out.

But in the late 1990s and early 2000s, a small group of cosmologists pointed out a major undiscussed assumption their colleagues were putting into their calculations. It was a perfectly reasonable assumption to make, because neglecting it would violate certain long-held principles of theoretical physics so fundamental that no one wanted to upset them. But these principles weren't required by the data, and in the end, as scientists, our first loyalty has to be to the data. Even if it means rewriting the fate of the universe.

OFF THE EDGE OF THE MAP

The simple question physicist Robert Caldwell and his colleagues asked was: what if w is less than -1? Say, -1.5? Or -2? Up until this point, such a possibility was generally thought too outlandish to be considered. Plots in papers showing the "allowed" region for w based on the data tended to abruptly cut off at -1. The axis might go from -1 to 0, or -1 to 0.5, but -1 was a hard wall, the same way you might put a hard wall at 0 when guessing a person's height.

But when Caldwell looked at the problem, all the observations of w pointed to a value of -1 or something very close to it. Which suggested that there might be values below -1 that were also allowed by the data, if only someone were to check. This hypothetical dark energy with w less than -1 was dubbed by Caldwell "phantom dark energy" and would be deeply inconsistent with the aforementioned Important Theoretical Principles—specifically, the "dominant energy condition," which says, roughly, that energy can't flow faster than light.[fn2] This seems like a completely sensible condition to place on the universe, but it's subtly different than the usual statement that light (or any kind of matter) has an ultimate speed limit, and it's currently less of a proven physical principle than a Very Good Idea. Maybe it's flexible?